The University of Faisalabad

LAB: 05

Topic: Implementation of Clauses in Database

```
create database university;
use university;
CREATE TABLE students (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(50),
    age INT,
    grade CHAR(1),
    city VARCHAR(50)
);
INSERT INTO students (id,name, age, grade, city) VALUES
(1, 'Alice', 20, 'A', 'New York'),
(2, 'Bob', 22, 'B', 'Los Angeles'),
(3, 'Charlie', 21, 'A', 'Chicago'),
(4, 'Diana', 23, 'C', 'Houston'),
(5, 'Eve', 20, 'B', 'New York');
SELECT * FROM students;
SELECT name, grade FROM students;
SELECT * FROM students WHERE city = 'New York';
SELECT * FROM students ORDER BY age DESC;
```

```
SELECT * FROM students LIMIT 3;

TRUNCATE TABLE students;

Select*from university.students;
```

Task 1: Data Retrieval and Filtering

1. Retrieve the names and grades of students who have an 'A' grade.
2. Display all students from the database, but only show those who are older than 21.
3. List students in ascending order of their names.
4. Fetch only the first two records from the table.

Task 2: Database Modification and Cleanup

1. Insert two new students into the **students** table with appropriate values.
2. Update the grade of a student named 'Eve' to 'A'.
3. Delete the record of a student from 'Los Angeles'.
4. Remove all records from the **students** table without deleting its structure.